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Variable-length analog of Stavskaya process: A new example of misleading simulation

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This article presents a new example intended to showcase limitations of computer simulations in the study of random processes with local interaction. For this purpose, we examine a new version of the well-known Stavskaya process, which is a discrete-time analog of the well-known contact processes. Like the bulk of random processes studied till now, the Stavskaya process is constant-length, that is, its components do not appear or disappear in the course of its functioning. The process, which we study here and call Variable Stavskaya, VS, is similar to Stavskaya; it is discrete-time; its states are bi-infinite sequences, whose terms take only two values (denoted here as "minus" and "plus"), and the measure concentrated in the configuration "all pluses" is invariant. However, it is a variable length, which means that its components, also called particles, may appear and disappear under its action. The operator VS is a composition of the following two operators. The first operator, called "birth," depends on a real parameter β; it creates a new component in the state "plus" between every two neighboring components with probability β independently from what happens at other places. The second operator, called "murder," depends on a real parameter α and acts in the following way: whenever a plus is a left neighbor of a minus, this plus disappears (as if murdered by that minus which is its right neighbor) with probability α independently from what happens to other particles. We prove for any $\alpha < 1$ and any $\beta > 0$ and any initial measure μ that the sequence $\mu(\mathsf{VS})^t$ (the result of t iterative applications of VS to μ) tends to the measure $\delta_\oplus$ (concentrated in "all pluses") as $t \to \infty$. Such a behavior is often called ergodic. However, the Monte Carlo simulations and mean-field approximations, which we performed, behaved as if $\mu(\mathsf{VS})^t$ tended to $\delta_\oplus$ much slower for some $\alpha,\ \beta,\ \mu$ than for some others. Based on these numerical results, we conjecture that VS has phases, but not in that simple sense as the classical Stavskaya process. *Published by AIP Publishing.* [http://dx.doi.org/10.1063/1.4983567]

I. INTRODUCTION

The goal of this article is to provide another example of discrepancy between the computer simulation of random processes and their theoretical study. Although some valuable studies have been conducted in this area (e.g., Refs. 1–13), we do not yet see a general theory. In this situation, we believe that further accumulation of diverse examples of such phenomena is a useful way of making progress in this important area of research.

The Stavskaya operator, Stav,[14] is a discrete-time analog of the well-known contact processes, see, e.g., Refs. 15 and 16. Stav has one parameter taking values in [0, 1]; if this parameter exceeds a certain critical value, which is greater than zero and less than one, the process has only one invariant measure concentrated in one configuration and tends to it from all initial conditions; in this case, we call the process ergodic; if this parameter is less than the critical value, the process fails to behave this way;[17] in this case, we call it non-ergodic. All the computer simulations of Stav, of which we are aware, comply with this theoretical result; even the mean-field approximation behaves in a qualitatively similar way, also with a critical value.

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Unlike that, the process, which we study here and call "Variable Stavskaya" or VS for short, is "variable-length," which means that its components, also called particles, may appear and disappear in the course of its functioning. Some classes of such processes have been studied.[18–24] A theoretical base for such studies was provided in Refs. 25 and 26. Like in the case with Stav, our process exhibits two tendencies: on the one hand pluses are born, but on the other hand they are murdered. Our goal is to understand which tendency will prevail.

We denote by $\mathbb{R}$ the set of real numbers, $\mathbb{Z}$ the set of integer numbers, and $\mathbb{Z}_+$ the set of natural numbers including zero. As it is usual in studies of this sort, we have a non-empty finite set called *alphabet*, whose elements are called *letters*. We call by a *word* any finite sequence of letters. The length of a word is the number of letters in it. Any letter may be treated as a word of length one. There is the empty word, denoted by Λ, whose length is zero. In this paper, we consider only one alphabet, which has two letters, called *minus* and *plus* and denoted by $\ominus$ and $\oplus$. For simplicity we may call by "minus" a component in a state "minus" and by "plus" a component in a state "plus." A particle means the same as a component. Our configuration space is $\{\ominus,\oplus\}^{\mathbb{Z}}$, the set of bi-infinite sequences, whose terms are minuses and pluses. We call these sequences *configurations*. We call a *thin cylinder* any set of the form

$$\left\{s \in \{\ominus,\oplus\}^{\mathbb{Z}} : s_i = a_i \text{ for all } i \in [m,n]\right\},$$

where s_i are components of s, i.e., variables, whose values are minuses and pluses and $a_i \in \{\ominus,\oplus\}$ are constants. We consider probability measures, that is, normalized measures on $\{\ominus,\oplus\}^{Z}$, that is, on the σ-algebra generated by thin cylinders. We denote by $\mathcal{M}$ the set of normalized measures on this σ-algebra. Convergence in $\mathcal{M}$ means convergence on all the thin cylinders. In this article, we use the following convention: measure $\mathsf{P}_1 \circ \cdots \circ \mathsf{P}_n$ event, that is, we write operators on the right side of measures in the order they are applied, and we write events on the right side of operators; also $\mu\mathsf{P}^t$ means the result of t iterative applications of the operator to the initial measure. Ends of arguments are marked by $\boxtimes$. Till now we called all maps from $\mathcal{M}$ to $\mathcal{M}$ operators; now we rename them into *interaction operators*. In particular, Stav and VS are interaction operators.

In this article, we deal only with a special class of interaction operators, known as *substitution operators*, defined in Refs. 25 and 26 and denoted by $V \xrightarrow{\rho} W$, where V, W are words and $\rho \in [0,1]$ is a real number. Informally speaking, this formula represents an interaction operator, which substitutes any word V in a configuration by a word W with a probability ρ (or leaves V unchanged with a probability $1-\rho$).

As usual, we define *translations* on $\mathbb{Z}$, translations on $\{\ominus,\oplus\}^{\mathbb{Z}}$, and then translations on $\mathcal{M}$. We call a measure *uniform* if it is invariant under all translations. The theory of substitution operators, developed till now, can deal only with uniform measures, so we concentrate our attention on them. This allows us to denote thin cylinders in the following simplified manner. For any uniform measure $\mu \in \mathcal{M}$ the formula $\mu(x_0, x_1, \ldots, x_n)$ means $\mu(x_i = a_0,\ x_{i+1} = a_1, \ldots, x_{i+n} = a_n)$ with arbitrary $i \in \mathbb{Z}$ and is called the *density* of the word $(x_0, x_1, \ldots, x_n)$ in measure μ. Our newly introduced operator VS has two real parameters α and β, both in [0, 1] (see formal definition in Section II). In this paper, we prove that for any real $\alpha < 1$ and $\beta > 0$ and any initial uniform measure $\mu \in \mathcal{M}$, the measure $\mu(\mathsf{VS})^t$ (the result of t iterative applications of VS to the initial measure μ) tends to one and the same measure $\delta_\oplus$ concentrated in the configuration "all pluses" as $t \to \infty$.

On the other hand, we performed mean-field approximations and Monte Carlo simulations with various values of parameters. In the mean-field approximation, the resulting sequences $\mu(\mathsf{VS})^t$ did not always tend to $\delta_\oplus$ and in the Monte Carlo simulations they did not always seem to tend to $\delta_\oplus$. Based on these approximations and numerical results, we conjecture that $\mu(\mathsf{VS})^t$ may tend to $\delta_\oplus$ in qualitatively different ways: sometimes faster and sometimes slower, and there is a phase transition between these two cases.

II. THE MAIN CLAIM

The substitution operator $\Lambda \xrightarrow{\beta} \oplus$, which we call *birth* and denote by $\mathsf{Birth}_{\mathbf{inter}} : \mathcal{M} \to \mathcal{M}$ (where **inter** means interaction), depends on a real parameter β. Under its action, between any two neighbor particles there may appear a new particle in the state "plus" with probability β independently from what happens at other places. If we considered a similar process on finite configurations, every act of birth of a plus would increase the length of the configuration by one.

The substitution operator $\oplus\ominus \xrightarrow{\alpha} \ominus$, which we call *murder* and denote by $\mathsf{Murder}_{\mathbf{inter}} : \mathcal{M} \to \mathcal{M}$, depends on a real parameter α. Under its action, whenever a plus is on the left side of a minus, this plus disappears with probability α (or remains unchanged with probability $1-\alpha$) independently from what happens to all the other particles. If we considered a similar process on finite configurations, every act of murder would decrease the length of the configuration by one.

Finally, we define the interaction operator VS as this composition,

$$\mathsf{VS} = \mathsf{Birth}_{\mathbf{inter}} \circ \mathsf{Murder}_{\mathbf{inter}}$$

(first $\mathsf{Birth}_{\mathbf{inter}}$ and then $\mathsf{Murder}_{\mathbf{inter}}$). In this and other cases, the sign $\circ$ means composition and may be omitted. Our main goal is to prove the following **main claim**:

$$\forall\ \mu \in \mathcal{M} : \mu\,(\,\mathsf{VS}\,)^t \to \delta_{\oplus} \text{ as } t \to \infty \tag{1}$$

for any $\alpha < 1$, any $\beta > 0$, and any initial measure $\mu \in \mathcal{M}$. (1) is equivalent to the following condition:

$$\forall\ \mu \in \mathcal{M} : \mu(\,\mathsf{VS}\,)^t(\oplus) \to 1 \text{ as } t \to \infty. \tag{2}$$

So we may prove (2) instead of (1).

Now we want to reduce the non-linear operator VS to a linear operator Single. But first we introduce some measures and some linear operators acting on them.

Notice that $\delta_{\oplus}$ is invariant for VS. Therefore, if the initial measure μ is $\delta_{\oplus}$, then our Main Claim (1) is trivially true. So we exclude this case and assume that the density of minus in μ is positive. Then the set of minuses in the initial configuration a.s. is a bi-infinite sequence and we may enumerate terms of this sequence by integer numbers. Since minuses never appear or disappear in our process, we may assume that minuses always keep the same space coordinates, that is, integer numbers.

We do not need to attribute exact space coordinates to pluses; it is sufficient to indicate to which space segment $(i, i+1)$ each of them belongs, assuming that whenever a new particle is born between two particles, its space position is between their positions. Since we consider only uniform measures, we may restrict our attention to the number of pluses (i.e., particles in the state plus) in the segment $(0, 1)$. As time t goes on, this number changes at every time step according to the $\mathcal{N}$-operator Single. Therefore at time t this number is distributed as $\mu_0(\mathsf{Single})^t$, where μ_0 is the initial distribution of the number of pluses in $(0, 1)$.

III. $\mathcal{N}$-MEASURES

We denote by $\mathcal{N}$ the set of probability measures on $\mathbb{Z}_+$. Elements of $\mathcal{N}$ are called $\mathcal{N}$-measures.

Let us say that a set $S \subset \mathbb{Z}_+$ is *upper* if

$$(x \in S, \quad x < y) \Longrightarrow y \in S.$$

Analogously we say that a set $S \subset \mathbb{Z}_+$ is *lower* if

$$(x \in S, \quad y < x) \Longrightarrow y \in S.$$

For any $\mathcal{N}$-measures $\mu,\ \nu$, we say that μ *precedes* ν and write $\mu \prec \nu$ if $\mu(S) \le \nu(S)$ for all upper S (which is equivalent to $\mu(S) \ge \nu(S)$ for all lower S). For any natural n we denote by δ_n the $\mathcal{N}$-measure concentrated in the state n. In particular, δ_0, concentrated in zero, precedes all $\mathcal{N}$-measures.

A $\mathcal{N}$-measure μ is called *local* if the set $\{x \mid \mu(x) \ne 0\}$ is finite. We call a pair $(\mu,\ \nu)$ *local* if both μ and ν are local. We call a local pair (μ, ν) a *step* if there are natural numbers a and b and a real number $p > 0$ such that $a < b$ and

$$\forall\ k \in \mathbb{Z}_+ : \nu(k) - \mu(k) = \begin{cases} -p, & \text{if } k = a, \\ \ \ p, & \text{if } k = b, \\ \ \ 0, & \text{for all the other values of } k. \end{cases}$$

Lemma 1. If (μ, ν) is a step, then $\mu \prec \nu$.

Proof. is easy. ⊠

Given two measures $\mu, \nu \in \mathcal{N}$, we call a sequence of measures $\pi_1, \ldots, \pi_n$ a *stairs* from μ to ν if $\pi_1 = \mu$, $\pi_n = \nu$ and every pair (π_i, π_{i+1}), where $i = 1, \ldots, n-1$, is a step. The case $\mu = \nu$ is included here with $n = 1$.

Lemma 2. Let us have a local pair (μ, ν). Then $\mu \prec \nu$ if and only if there is a stairs from μ to ν.

Proof. In one direction ($\Leftarrow$) it follows from Lemma 1. Let us prove it in the other direction ($\Rightarrow$). This is trivial in the case $\mu = \nu$, so we exclude it. Then for any local pair (μ, ν) we denote by $\max(\mu, \nu)$ and call the *maximum* of the pair (μ, ν) the greatest natural m such that $\mu(m) \neq 0$ or $\nu(m) \neq 0$. We shall prove Lemma 2 by induction in $\max(\mu, \nu)$. *Base of induction.* Suppose that $\max(\mu, \nu) = 0$. Then $\mu = \nu$ and Lemma 2 is true in this case.

Induction step. Suppose that Lemma 2 is already proved for all pairs, whose maximum does not exceed m. Suppose that now we have a new local pair (μ, ν), whose maximum equals $m + 1$ and such that $\mu \prec \nu$. We shall subject this pair to the following operation.

Since the set $[m+1, \infty)$ is upper, $\mu[m+1, \infty) \leq \nu[m+1, \infty)$. On the other hand, since $\max(\mu, \nu) = m + 1$,

$$\mu[m + 1, \infty) = \mu(m + 1) \text{ and } \nu[m + 1, \infty) = \nu(m + 1).$$

Therefore $\mu(m + 1) \leq \nu(m + 1)$. Now let us consider two cases.

Case 1. Suppose that $\mu(m + 1) > 0$. Then we perform the following operation: we subtract from both values $\mu(m + 1)$, $\nu(m + 1)$ one and the same quantity, namely, $\mu(m + 1)$. The stairs, which we know to exist for the resulting pair, is valid for the pair (μ, ν) also.

Case 2. Suppose that $\mu(m + 1) = 0$. Then $\nu(m + 1) > 0$. Then there exists $k < m + 1$ such that $\mu(k) > \nu(k)$. Then we take

$$p = \min\left(\mu(k) - \nu(k),\ \nu(m + 1) - \mu(m + 1)\right)$$

and form the following step:

$$\forall\ i \in \mathbb{Z}_+ : \mathsf{step}(k) = \begin{cases} -p, & \text{if } i = k, \\ p, & \text{if } i = m + 1, \\ 0, & \text{in all the other cases.} \end{cases} \tag{3}$$

Maximum of the resulting pair (μ', ν') does not exceed m. Hence, by the induction hypothesis this pair is connected by a stairs. Therefore also the pair (μ, ν) is connected by stairs obtained from it by including one extra step (3). *Lemma 2 is proved completely.* ⊠

For any $\mathcal{N}$-measure μ we define *expectation*

$$\mathbb{E}(\mu) = \sum_{i=0}^{\infty} i \cdot \mu(i).$$

Lemma 3.

(a) Any local $\mathcal{N}$-measure has a finite expectation.
(b) If (μ, ν) is a local step, then $\mathbb{E}(\mu) \leq \mathbb{E}(\nu)$.
(c) If μ and ν are local $\mathcal{N}$-measures, then $\mu \prec \nu \Longrightarrow \mathbb{E}(\mu) \leq \mathbb{E}(\nu)$.

Proof. Items (a) and (b) are easy; (c) follows from (b)

⊠

IV. $\mathcal{N}$-OPERATORS

Starting here, we apply the term $\mathcal{N}$*-operator* to time-homogeneous discrete time Markov chains with the set of states $\mathbb{Z}_+$. Thus any $\mathcal{N}$-operator is a Markov chain acting on $\mathcal{N}$, whose matrix elements are denoted by $\mu(y \mid x)$: the conditional probability to go to y if we are in x.

The relation $\prec$ defines a partial order on the set of measures on $\mathbb{Z}_+$. So, it is natural to obtain conditions for which a map preserves this order. Such maps are known as *order-preserving mappings* (see p. 21 in Ref. 27). We call an $\mathcal{N}$-operator P *local* if for every local measure μ on $\mathcal{N}$, the measure μP is also local. An $\mathcal{N}$-operator P is called *monotonic* if

$$\forall \;\; \mu, \nu : (\mu \prec \nu) \Longrightarrow (\mu P \prec \nu P).$$

Lemma 4. Let P be a local $\mathcal{N}$-operator. Then

$$P \text{ is monotonic} \Longleftrightarrow (\forall \;\; m, n \in \mathbb{Z}_+ : m < n \Longrightarrow \delta_m P \prec \delta_n P\,)\,.$$

Proof. In one direction ($\Rightarrow$) Lemma 4 is evident. In the other direction ($\Leftarrow$) it follows from Lemmas 1 and 2. ⊠

Given $\mathcal{N}$-operators P, Q, we say that P precedes Q and write $P \prec Q$ if $\mu\, P \prec \mu\, Q$ for all $\mu \in \mathcal{N}$.

Lemma 5. Let us have monotonic $\mathcal{N}$-operators P_1, P_2, Q such that $P_1 \prec P_2$. Then $P_1\, Q \prec P_2\, Q$ and $QP_1 \prec QP_2$.

Proof. is *easy*. ⊠

Lemma 6. Given local monotonic $\mathcal{N}$-operators $P_1, \ldots, P_n$ and $Q_1, \ldots, Q_n$ such that $P_1 \prec Q_1, \ldots, P_n \prec Q_n$. Then

$$P_1 \circ P_2 \circ \cdots \circ P_n \;\prec\; Q_1 \circ Q_2 \circ \cdots \circ Q_n.$$

Proof. By induction. *Base of induction:* $P_1 \prec Q_1$ is given.

Induction hypothesis: $P_1 \circ \cdots \circ P_{n-1} \prec Q_1 \circ \cdots \circ Q_{n-1}$.

Induction step: It is clear that

$$P_n \circ (P_1 \circ \cdots \circ P_{n-1}) \prec Q_n \circ (P_1 \circ \cdots \circ P_{n-1}). \tag{4}$$

Also, we get that $(P_1 \circ \cdots \circ P_{n-1})$ is a monotonic $\mathcal{N}$-operators. Thus, using Lemma 5 we can rewrite (4) as follows:

$$(P_1 \circ \cdots \circ P_{n-1}) \circ P_n \prec (P_1 \circ \cdots \circ P_{n-1}) \circ Q_n. \tag{5}$$

Now, from the induction hypothesis and monotonicity of Q_n

$$(P_1 \circ \cdots \circ P_{n-1}) \circ Q_n \prec (Q_1 \circ \cdots \circ Q_{n-1}) \circ Q_n. \tag{6}$$

Now Lemma 6 follows from (5) and (6). ⊠

V. FROM NON-LINEAR, VS TO LINEAR SINGLE

It is well-known that most results in the area of random processes with local interaction speak about linear operators. Unlike that, substitution operators are generally non-linear, which makes them harder to manage. (In a non-linear case, even such an elementary property as transitivity may take an involved argument, see, e.g., Ref. 28.) However, the present case is special: VS, although non-linear, may be reduced to a bi-infinite sequence of random processes, each determined by a linear operator Single. Let us describe it.

First for every natural i we define a random variable B_i, which is a sum of i independent random variables, everyone of which equals 1 with probability β and 0 with probability $(1 - \beta)$.

Now we use the formula $y = x + B_{x+1}$ to define a $\mathcal{N}$-operator, which we denote by Birth because it corresponds to the action of $\mathsf{Birth}_{\mathbf{inter}}$. The same operator may be defined by the formula

$$\text{For all } x,\ y \in \mathbb{Z}_+ : \mathsf{Birth}(y \mid x) = \begin{cases} \binom{x+1}{y-x} \cdot \beta^{y-x} \cdot (1-\beta)^{2x-y+1}, & \text{if } x \le y \le 2x+1, \\ 0, & \text{in all the other cases.} \end{cases} \tag{7}$$

Another $\mathcal{N}$-operator, which we denote by Murder because it corresponds to the action of $\mathsf{Murder}_{\mathbf{inter}}$, acts as follows:

$$\mathsf{Murder}\,(y \mid x) = \begin{cases} \alpha, & \text{if} \quad x > 0 \quad \text{and} \quad y = x-1, \\ 1-\alpha, & \text{if} \quad x > 0 \quad \text{and} \quad y = x, \\ \alpha, & \text{if} \quad x = 0 \quad \text{and} \quad y = 0, \\ 1-\alpha, & \text{if} \quad x = 0 \quad \text{and} \quad y = 1, \\ 0, & \text{in all the other cases.} \end{cases} \tag{8}$$

Finally we define the operator Single as a composition

$$\mathsf{Single} = \mathsf{Birth} \circ \mathsf{Murder} \tag{9}$$

(first Birth and then Murder). Thus we have defined the $\mathcal{N}$-operators Birth, Murder, and Single.

Lemma 7. The $\mathcal{N}$-operators Birth, Murder, *and* Single *are local and monotonic.*

Proof. Locality of Birth and Murder is evident. Their monotonicity follows from Lemma 4. Hence the same is true for Single because a composition of local and monotonic $\mathcal{N}$-operators is local and monotonic. ⊠

Let μ_0 be the initial distribution of the number of pluses in (0, 1). We assume that $\mu_0 = \delta_0$, that is, initially the number of pluses in (0, 1) is zero. In this case, we may write an equality which claims that the proportion of pluses to minuses in the whole line is the same as in the segment [0, 1),

$$\delta_{\ominus}(\mathsf{VS})^t(\oplus) = \frac{\mathbb{E}(\delta_0(\mathsf{Single})^t)}{1 + \mathbb{E}(\delta_0(\mathsf{Single})^t)}, \tag{10}$$

where $\mathbb{E}(\cdot)$ means mathematical expectation. In Secs. VI–VIII we shall prove that

$$\mathbb{E}(\delta_0(\mathsf{Single})^t) \to \infty \text{ as } t \to \infty. \tag{11}$$

Together with (10) this implies (2) in the special case $\mu = \delta_{\ominus}$.

Now, assuming (11), let us take care of the arbitrary initial distribution. From Lemmas 6 and 7

$$\forall\ \mu',\ \mu \in \mathcal{N} : \mu' \prec \mu \Longrightarrow \mu'(\mathsf{Single})^t \prec \mu(\mathsf{Single})^t$$

for all $\mu',\ \mu \in \mathcal{N}$ and all natural t. Setting here $\mu' = \delta_0$, we get

$$\forall\ \mu \in \mathcal{N} : \delta_0(\mathsf{Single})^t \prec \mu(\mathsf{Single})^t. \tag{12}$$

Hence from Lemma 3

$$\mathbb{E}(\delta_0(\mathsf{Single})^t) \le \mathbb{E}(\mu(\mathsf{Single})^t).$$

Hence and from (11) we get

$$\mathbb{E}(\mu(\mathsf{Single})^t) \to \infty \text{ as } t \to \infty \tag{13}$$

for any initial measure μ. Thus we only need to prove (11).

VI. COMPLETE GROWTH

Let us say that a sequence of real random variables (X_k) *completely grows* as $k \to \infty$ if for every real number x the sequence of numbers $\mathbb{P}(X_k \le x)$ tends to zero as $k \to \infty$.

Lemma 8. If a sequence (X_k) of random variables on $\mathbb{Z}_+$ completely grows as $k \to \infty$, then the sequence of their mathematical expectations $\mathbb{E}(X_k)$ tends to ∞ as $k \to \infty$.

Proof. Since the sequence (X_k) completely grows, for any N and any $\varepsilon > 0$ there is k_0 such that

$$\forall\, k \geq k_0 : \mathbb{P}(X_k \geq N) \geq 1 - \varepsilon.$$

It is well-known that for any random variable X and any number N'

$$\mathbb{E}(X) \geq N' \cdot \mathbb{P}(X \geq N').$$

Choosing $\varepsilon = 1/2$ and $N' = 2N$, we obtain

$$\forall\, k \geq k_0 : \mathbb{E}(X_k) \geq N' \cdot \mathbb{P}(X_k \geq N') \geq N' \cdot (1 - \varepsilon) = 2N \cdot \frac{1}{2} = N.$$

Lemma 8 is proved.

Due to Lemma 8, instead of (11) it is sufficient to prove that

$$\textit{the sequence } \ \delta_0 (\mathsf{Single})^t \ \textit{ completely grows as } t \to \infty. \tag{14}$$

We shall prove (14) in Section IX.

VII. $\mathcal{N}$-OPERATORS SING AND SIN$_r$

Let us define another local $\mathcal{N}$-operator Sing, whose behavior is qualitatively similar to that of Single, but simpler,

$$\mathsf{Sing}\,(y|x) = \begin{cases} 2/3, & \text{if } q \leq x \text{ and } y = x + 1, \\ 1/3, & \text{if } q \leq x \text{ and } y = x - 1, \\ \gamma, & \text{if } x < q \text{ and } y = x + 1, \\ 1 - \gamma, & \text{if } 0 < x < q \text{ and } y = x - 1, \\ 1 - \gamma, & \text{if } x = y = 0, \\ 0, & \text{in all the other cases.} \end{cases}$$

Here the natural number q and the positive number γ are parameters, which we shall choose in an appropriate way. Notice that $\mathbb{P}(B_x \leq 1)$ tends to zero as $x \to \infty$ for any $\beta > 0$. Using this,

$$\left.\begin{array}{c} \textit{we choose } q \textit{ as the smallest natural number such that} \\ \mathbb{P}(B_{x+1} \leq 1) \leq 1/3 \text{ for all } x \geq q. \end{array}\right\} \tag{15}$$

Having chosen q, we define γ as follows:

$$\gamma = \min\left(\beta\,(1 - \alpha),\ \min\left\{ \mathbb{P}\,(B_{x+1} > 1) \,\middle|\, 0 < x < q \right\} \right). \tag{16}$$

Thus Sing is defined.

Now we want to consider some finite Markov chains. First we define an operator $\mathsf{Cut}_r : \mathbb{Z}_+ \to \mathbb{Z}_+$: which turns any $x \in \mathbb{Z}_+$ into $\min(x, r)$. After that for any natural r we define Sin_r as a composition,

$$\mathsf{Sin}_r = \mathsf{Sing} \circ \mathsf{Cut}_r.$$

Lemma 9. For q and γ defined in (15) and (16), respectively, and for any $r > q$,

(a) Sin_r, Sing, *and* Single *are local and monotonic and*
(b) $\mathsf{Sin}_r \prec \mathsf{Sing} \prec \mathsf{Single}$.

Proof is evident. ⊠

VIII. OPERATORS SIN'_r AND THEIR INVARIANT MEASURES

The operator Sin_r acts on the whole of $\mathbb{Z}_+$, but we are interested only in the sites $\{0, \ldots, r\}$ now; so we consider a finite Markov chain with the set of states $\{0, \ldots, r\}$, whose transition probabilities within $[0, r]$ are the same as that of Sin_r and denote it by Sin'_r. Thus Sin'_r is a finite irreducible Markov chain. So it has exactly one normalized invariant measure, which we denote by λ_r.

Lemma 10. The sequence (λ_r) completely grows as $r \to \infty$.

Proof. We shall prove Lemma 10 by writing explicitly the measure λ_r. We can represent Sin'_r as a matrix, which we shall denote by $\mathbf{M}$. So λ_r is the normalized solution of the equation

$$\lambda_r(\mathbf{M} - \mathbf{I}) = \mathbf{0}, \tag{17}$$

where $\mathbf{I}$ is the matrix identity of order $r + 1$ and $\mathbf{0}$ is the null matrix with $r + 1$ columns.

Notice that the only non-zero transition probabilities of Sin'_r are those which go from any s to $s - 1$, s, or $s + 1$. This allows us to use Lemma 11 to conclude that the invariant measure λ_r of Sin'_r satisfies the following conditions:

$$\left.\begin{aligned} \gamma \cdot \lambda_r(s) &= (1-\gamma)\cdot\lambda_r(s+1) \text{ for } s = 0, \ldots, q-2, \\ 3\gamma \cdot \lambda_r(q-1) &= \lambda_r(q), \\ 2 \cdot \lambda_r(s) &= \lambda_r(s+1) \text{ for } s = q, \ldots, r-1. \end{aligned}\right\} \tag{18}$$

From (18), for all $s \in [0, r]$

$$\lambda_r(s) = \begin{cases} \lambda_r(q) \cdot \dfrac{1}{3\gamma} \cdot \left(\dfrac{1-\gamma}{\gamma}\right)^{q-s}, & \text{for } 0 \le s < q, \\ \lambda_r(q) \cdot 2^{s-q}, & \text{for } q \le s \le r \end{cases},$$

where $\lambda_r(q)$ is a parameter which we yet need to choose. Using the formula of a sum of geometrical progression, we get

$$\sum_{s=q}^{r} \lambda_r(s) = \lambda_r(q) \cdot (2^{r-q+1} - 1).$$

To calculate the sum of $\lambda_r(s)$ from $s = 0$ to $s = q - 1$, it is better to enumerate the states in the inverse order. Then we have a geometric progression with q terms, in which the first term is

$$\lambda_r(q-1) = \frac{1}{3\gamma} \cdot \frac{1-\gamma}{\gamma} \cdot \lambda_r(q)$$

and the coefficient is $\frac{1-\gamma}{\gamma}$. Thus

$$\sum_{s=0}^{q-1} \lambda_r(s) = \frac{1}{3\gamma} \cdot \left(1 + \cdots + \phi^{q-1}\right) = \frac{1}{3\gamma} \cdot \frac{\phi(\phi^q - 1)}{\phi - 1},$$

where $\phi = \dfrac{1-\gamma}{\gamma}$. Also we denote $\Phi = \dfrac{1}{3\gamma} \cdot \dfrac{\phi(\phi^q - 1)}{\phi - 1}$.

Then $\displaystyle\sum_{s=0}^{r} \lambda_r(s) = \lambda_r(q) \cdot \left(\Phi + (2^{r-q+1} - 1)\right)$.

To normalise the measure λ_r, we take

$$\lambda_r(q) = \frac{1}{\Phi + (2^{r-q+1} - 1)}.$$

Therefore for any $r' \in [q, r]$

$$\sum_{s=0}^{r'} \lambda_r(s) = \frac{\Phi + (2^{r'-q+1} - 1)}{\Phi + (2^{r-q+1} - 1)}.$$

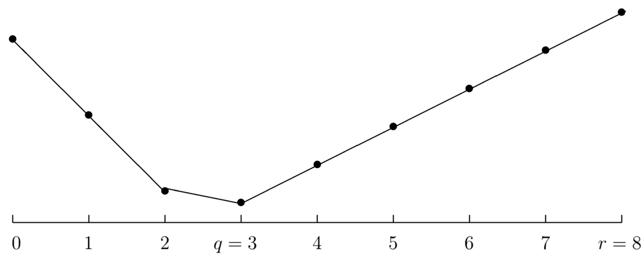

FIG. 1. The black circles show values of λ_r, the normalized invariant measure of the operator Sin_r, for the case q = 3 and r = 8 in the logarithmic scale. The lines connecting them are just to assist visual impression. The left part of the graph (from zero to q − 1) represents states in which the process (that started at zero) spends long time to reach the state q, but once the state q is reached, the process easily grows to r.

The value of this fraction tends to zero as r tends to ∞, all the other parameters including r' kept fixed. Lemma 10 is proved.

Now to prove (14). From Lemma 10

$$\forall\, r_1 : \lambda_r[0,\ r_1] \to 0 \text{ as } r \to \infty,$$

where $[0,\ r_1]$ is the segment with the ends 0 and r_1. The same in more detail gives

$$\forall\, r_1\ \forall\, \varepsilon > 0\ \ \exists\, r_0\ \forall\, r \geq r_0 : (\lambda_r[0,\ r_1] < \varepsilon/2\,). \tag{19}$$

On the other hand, since Sin'_r is irreducible,

$$\delta_0\ (\mathsf{Sin}'_r)^t \to \lambda_r \text{ as } t \to \infty.$$

In particular

$$\forall\, r_1 : \delta_0\ (\mathsf{Sin}'_r)^t\ [0,\ r_1] \to \lambda_r\ [0,\ r_1] \text{ as } t \to \infty.$$

In more detail

$$\forall\, r_1,\ \forall\, \varepsilon > 0\ \ \exists\, r_0, \exists\, t_0\ \forall\, r \geq r_0, \forall\, t \geq t_0\ :$$

$$|\ \delta_0\ (\mathsf{Sin}'_r)^t\ [0,\ r_1] - \lambda_r[0,\ r_1]\ |\ \leq \varepsilon/2. \tag{20}$$

Now, combining (*19*) and (*20*), *we get*

$$\forall\, r_1\ \forall\, \varepsilon > 0\ \ \exists\, r_0\ \exists\ \ t_0\ \forall\, r \geq r_0,\ \forall\, t \geq t_0\ :\ \delta_0(\mathsf{Sin}'_r)^t\ [0,\ r_1] \leq \varepsilon.$$

From Lemma 9

$$\forall\, r_1\ \forall\, \varepsilon > 0\ \ \exists\ \ t_0\ \ \forall\, t \geq t_0 :\ \delta_0(\mathsf{Single})^t\ [0,\ r_1] \leq \varepsilon.$$

Hence it follows (14), which implies formula (1), our main claim.

The illustration of the operator Sin_r, given by Figure 1, presents an apparent connection with the nucleation of a critical Wulff droplet,[29] probability of nucleation of the critical droplets appears to play a similar role, related to the time that is necessary to reach the state q, but once it got there, it grows fast to reach r.

IX. MEAN-FIELD APPROXIMATION

Now we go to numerical studies of the same process $\mu(\mathsf{VS})^t$. We start with the mean-field approximation. The difference between its behavior and the behavior of VS is not astonishing, but its advantage is that its behavior can be described explicitly. We denote by $C\ :\ \mathcal{M} \to \mathcal{M}$ the well-known *mean-field operator.*[18,20] Its action amounts to mixing randomly all the components. In other words, for each $\mu \in \mathcal{M}$ the measure μC is a product-measure with the same densities of all the letters as μ has. (This method is also known as the mean-field approximation.) The mean-field operator allows us to approximate a given process $\mu\,(P)^t$ on the configuration space $\{\ominus, \oplus\}^{\mathbb{Z}}$ by another process $\mu\,(C\,P)^t$ on the same space. (Every time, we apply first C and then P.) Thus, instead of the original process, whose set of parameters is infinite or very large, we deal with the evolution of densities of letters, that is, a finite and limited set of parameters. Since densities of the letters sum up to one, the number of independent parameters in the mean-field approximation equals the number of letters in the alphabet minus one. In our case, with only two letters, we deal with only one parameter: as such we choose the density of pluses.

Due to the properties of the mean-field operator, the density of pluses in the measure μC Birth Murder depends only on the density of pluses in the measure μ and this dependence may be expressed by the formula

$$f(x) = \frac{b - \alpha b(1-b)}{1 - \alpha\, b(1-b)}, \tag{21}$$

where x denotes the density of pluses in the measure μ, $f(x)$ denotes the density of pluses in the measure μC Birth Murder, and $b = (x+\beta)/(1+\beta)$. (b is the density of pluses in the measure μC Birth.) We consider only the case $\alpha\, b < 1$ because the case $\alpha\, b = 1$ implies $x = 1$ and $\alpha = 1$, which is trivial. So $f(t)$ is defined and continuous for $x,\ \alpha,\ \beta \in [0,\ 1]$. Thus the study of the operator Birth $\circ$ Murder is substituted by a study of the operator $C \circ$ Birth $\circ$ Murder, which boils down to the study of the dynamical system $f : [0, 1] \to [0, 1]$ with parameters $\alpha,\ \beta \in [0,\ 1]$. As usual, we call a *fixed point* of f any value of $x \in [0,\ 1]$ such that $f(x) = x$. We call our dynamical system *ergodic* if it has a unique fixed point x_{fixed} and

$$\forall\, x \in [0,\ 1] : \lim_{t\to\infty} f^t(x) = x_{\text{fixed}},$$

where f^t means the tth iteration of f.

Proposition 1. The mean-field approximation is ergodic if $\beta > \beta^*(\alpha)$ and is not ergodic if $\beta \le \beta^*(\alpha)$, where

$$\beta^*(\alpha) = \begin{cases} \dfrac{\alpha}{4-\alpha}, & \text{if } \alpha > 0, \\ 0, & \text{if } \alpha = 0. \end{cases} \tag{22}$$

Thus for the mean-field approximation, we know exactly the boundary between ergodicity and non-ergodicity. The graph of this curve is labeled "M.F." in Figure 2.

Proof of Proposition 1. First let $\alpha = 0$. It is easy to observe that in this case, our dynamical system is ergodic for all $\beta > 0$ and non-ergodic for $\beta = 0$, so Proposition 1 is true. Now let $\alpha > 0$. If β equals zero or one, Proposition 1 is evident. So now we additionally assume that $0 < \beta < 1$. Solving the equation $f(x) = x$, we find out that all the fixed points of f are

$$p_1 = \frac{\alpha(1-\beta) - \sqrt{\delta}}{2\alpha}, \qquad p_2 = \frac{\alpha(1-\beta) + \sqrt{\delta}}{2\alpha}, \qquad p_3 = 1, \tag{23}$$

which are real and belong to [0, 1], where

$$\delta = \beta^2(\alpha^2 - 4\alpha) + \beta(2\alpha^2 - 4\alpha) + \alpha^2. \tag{24}$$

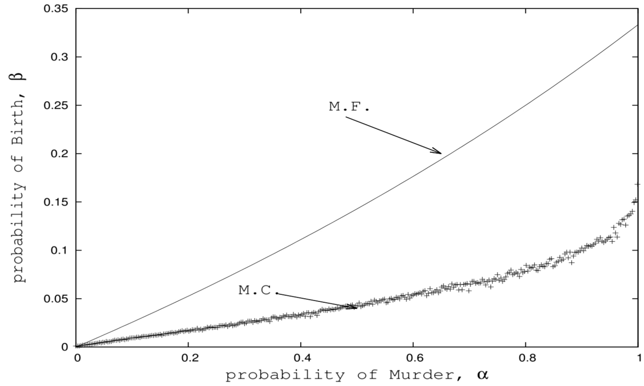

FIG. 2. This graph shows the curves obtained by Monte Carlo Simulation (M.C.) and mean-field approximation (M.F.). Each curve separates the region where the process seems to be ergodic from that where it seems to be non-ergodic according to each approximation.

Since $\alpha > 0$ and $\beta < 1$, the denominator of p_1 and p_2 in (23) is not zero. Since $\beta > 0$,

$$0 < p_1 \le p_2 < p_3 = 1 \tag{25}$$

whenever p_1 and p_2 are real, that is, $\delta \ge 0$. δ equals zero at $\beta = \beta_1(\alpha)$ and $\beta = \beta_2(\alpha)$, where

$$\beta_1(\alpha) = -1 \text{ and } \beta_2(\alpha) = \frac{\alpha}{4-\alpha}. \tag{26}$$

Here β_2 is the same as what we called β^*. According to (24), δ is a quadratic function of β with a negative second-degree coefficient, so δ is positive when β is between roots (26) and negative when β is less than $\beta_1(\alpha)$ or greater than $\beta_2(\alpha)$. It is easy to observe that

$$\forall\ \alpha > 0 : \beta_1(\alpha) < 0 < \beta_2(\alpha) < 1. \tag{27}$$

If $\beta_1(\alpha) < \beta < \beta_2(\alpha)$, $\delta > 0$ and our dynamical system has three different fixed points p_1, p_2, and p_3, so it is non-ergodic. If $\beta = \beta_1(\alpha)$ or $\beta = \beta_2(\alpha)$, $\delta = 0$ and our dynamical system has two different fixed points $p_1 = p_2$ and p_3, so it is non-ergodic too.

Now let $\beta < \beta_1(\alpha)$ or $\beta > \beta_2(\alpha)$. In this case $\delta < 0$, so f has only one fixed point $p_3 = 1$. Let us prove that for all $x_0 \in [0, 1]$ the sequence $f^t(x_0)$ tends to 1 when t tends to infinity. As we know, f is continuous in [0, 1]. So, $g(x) = f(x) - x$ is also continuous. It is easy to calculate that $g(0) > 0$. Since g is continuous and equals zero only at the point $x = 1$ and $g(0) = f(0) > 0$, we conclude that $g(x) > 0$ for all $x < 1$, whence $f(x) > x$ for all $x < 1$. Then for all $x_0 < 1$, the sequence $f^t(x_0)$ is growing and limited by 1 and therefore has a limit, which is a fixed point of f. But in this case the number 1 is the only fixed point of f, so $f^t(x_0) \to 1$ when $t \to \infty$. *Proposition 1 is proved completely.*

In fact, we can describe completely the limit behavior of this dynamical system,

$$\text{if } \delta < 0, \text{ then } \lim_{t\to\infty} f^t(x_0) = p_3 = 1 \text{ for all } x_0,$$

$$\text{if } \delta = 0, \text{ then } \lim_{t\to\infty} f^t(x_0) = \begin{cases} p_1 = p_2, & \text{if } x_0 \le p_1 = p_2, \\ p_3 = 1, & \text{if } x_0 > p_1 = p_2, \end{cases}$$

$$\text{if } \delta > 0, \text{ then } \lim_{t\to\infty} f^t(x_0) = \begin{cases} p_1, & \text{if } x_0 < p_2, \\ p_2, & \text{if } x_0 = p_2, \\ p_3 = 1, & \text{if } x_0 > p_2. \end{cases}$$

X. MONTE CARLO SIMULATION

Now we go to the Monte Carlo simulation, whose discrepancy with the rigorous study is really astonishing. Based on results of Section V, instead of simulating the process VS, we shall simulate the process Single, which is more easy. Let us describe a procedure, which we call *Imitation* and which is a Monte Carlo imitation of our process. This procedure generates a sequence of random variables in the following inductive way.

Base of induction: $Z_0 = 0$.

tth induction step is as follows:

- Fix α and β on the interval [0, 1].
- Given Z_{t+1}, where $t = 1, 2, 3, \ldots$ perform two procedures:

First procedure imitating Birth *is as follows:*

- Generate a variable B with binomial distribution, $bin(Z_t + 1, \beta)$.
- Set $Y = Z_t + B$.

Second procedure imitating Murder *is as follows:*

Generate a random variable U distributed uniformly in the interval [0, 1]. Thus, what occurs is one of the following three cases:

- $X = Y - 1$, if $U < \alpha$ and $Y > 0$;
- $X = Y$, if $U > \alpha$ and $Y > 0$;
- $X = Y$, if $Y = 0$.

Let $Z_{t+1} = X$.

When we stop: given a constant $T = 1\,000\,000$, we stop when $t = T$ or $Z_t > 10\,000$.

We use the above described numerical procedure to attribute the appropriate value to a Boolean variable denoted by E (which means ergodicity), namely, E is set as *yes* if Z_t is greater than 10 000; otherwise E is set as *no*. If E = *yes*, we interpret this as a suggestion that the process with the given values of α and β is ergodic; the result E = *no* is taken as a suggestion that our process is non-ergodic.

In fact we used imitation within a cycle with growing β: we started with $\beta = 0$ and then iteratively performed imitation and increased β by 0.001 and repeated this until β reached the value 1 or E got the value *yes*, that is, ergodicity was suggested. Thus we obtained a certain value of β. In fact, we performed this cycle 10 times and recorded the arithmetical average of the 10 values of β thus obtained.

Remember that all this was done with a certain value of α. In fact we considered 1000 values of α, namely, the values $\alpha_i = 0,001 \cdot i$ for $i = 1, \ldots, 1000$. The corresponding recorded value of β was denoted by β_i. Thus we obtained 1000 pairs (α_i, β_i). The graph called M.C. in Figure 2 consists of these pairs plotted.

Of course, what we call the "Monte Carlo curve" is not really a curve, it is somewhat fuzzy. But if we performed more than five procedures, this graph would be thinner. Indeed, even such as it is, it gives some idea of the behavior of our process.

Given a pair (α, β), we define

$$\max(Z_t) = \max\{Z_t :\ t = 0, \ldots, 100\,000\}.$$

If $\max(Z_t) \geq 10\,000$, we interpret the pair (α, β) as belonging to the area of ergodicity. Otherwise we conclude that the pair (α, β) is in the non-ergodicity area.

We studied in more detail the behavior of $\max(Z_t)$. We took several values of α and for everyone of them made 100 experiments, in each of them making β grow from zero to one with an increment 0.0001, all the time calculating the maximum value of Z_t, which we denoted by $\max(Z_t^i)$. Then we defined

$$\mathbb{E}[\max(Z_t^i)] = \frac{1}{100}\sum_{i=1}^{100} \max(Z_t^i). \tag{28}$$

The behavior of $\mathbb{E}[\max(Z_t^i)]$ for $\alpha = 0.5$ is shown in Figure 3. We observe that when β increases, $\mathbb{E}[\max(Z_t^i)]$ also increases and that near the transition area there is an abrupt increase. The same

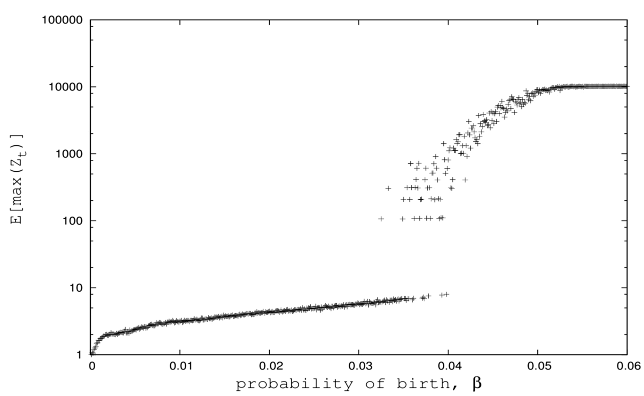

FIG. 3. Behavior of $\mathbb{E}(\max(Z_t^i))$ for $\alpha = 0.5$. It grows sharply near the critical value. For each value of β, we made 100 independent experiments.

experiment shows similar qualitative behavior of $\mathbb{E}[\max(Z_t^i)]$ for $\alpha = 0.1,\ 0.2,\ 0.3,\ 0.4, 0.6, 0.7, 0.8,$ 0.9.

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APPENDIX: A RESULT TO IRREDUCIBLE MARKOV CHAIN

1. AB-lemma

Lemma 11. Let us have a finite irreducible Markov chain with a set of states $S = A \cup B$, where $A \cap B$ is empty. For all $x,\ y \in S$ we denote by $p(y \mid x)$ the probability of transition from x to y. These probabilities form the transition matrix. It is well-known that under these conditions a Markov chain has a unique invariant measure, which we denote by μ. Let us fix two states $a \in A$ and $b \in B$ and assume that $p(y \mid x) = p(x \mid y) = 0$ for all $x \in A$ and $y \in B$ except the case $x = a$, $y = b$. Then

$$\mu(a) \cdot p(b \mid a) = \mu(b) \cdot p(a \mid b). \tag{A1}$$

Proof of Lemma 11. Of course

$$\forall\, x \in S : \mu(x) = \sum_{y \in S} \mu(y) \cdot p(x \mid y).$$

Therefore

$$\sum_{x \in A} \mu(x) = \sum_{x \in A} \sum_{y \in S} \mu(y) \cdot P(x \mid y) =$$
$$\sum_{x \in A} \sum_{y \in A} \mu(y) \cdot P(x \mid y) + \sum_{x \in A} \sum_{y \in B} \mu(y) \cdot P(x \mid y).$$

Here the second sum has only one non-zero term, namely, that one, which has $x = a$ and $y = b$. Therefore the second sum equals $\mu(b) \cdot p(a \mid b)$.

The first sum equals

$$\sum_{y \in A} \sum_{x \in A} \mu(y) \cdot p(x \mid y) =$$
$$\sum_{y \in A} \sum_{x \in S} \mu(y) \cdot p(x \mid y) - \sum_{y \in A} \sum_{x \in B} \mu(y) \cdot p(x \mid y).$$

Here the minuend equals

$$\sum_{y \in A} \mu(y) \sum_{x \in S} p(x \mid y) = \sum_{y \in A} \mu(y) = \sum_{x \in A} \mu(x).$$

The subtrahend has only one non-zero term, namely, that one, which has $y = a$ and $x = b$. Therefore it equals $\mu(a) \cdot p(b \mid a)$. Thus

$$\sum_{x \in A} \mu(x) = \sum_{x \in A} \mu(x) + \mu(b) \cdot p(a \mid b) - \mu(a) \cdot p(b \mid a).$$

Hence (A1) follows. ⊠

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